Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_30

1. Summary of Key Deviations

The analysis of Run_30 reveals a significant physical fault characterized by a system-wide loss of pressure, despite the pump and bypass working harder than normal.

- **p_Combustion (Moderate):** The most significant and consistent anomaly is a moderate **negative offset** on the combustion pressure, which reads approximately **-11%** lower than healthy across all active states.
- **Flow_Bypass (Moderate):** The bypass flow shows a large positive **offset** of approximately **+13%**.
- **Flow_Pump (Minor):** The pump flow also shows a significant positive **offset**, reading **+7.5%** higher than the healthy baseline.
- **Flow_PRV (Moderate):** Uniquely, the `Flow_PRV` signal shows a moderate positive **offset** of **+13.8%** at the very beginning of the run (state s1), indicating the high-pressure relief valve is active even at idle.
- **Other Signals (Nominal):** All other signals, including `Flow_Engine`, are either nominal or show only negligible deviations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Pressure Relief Valve (PRV)**	**Stuck Partially Open / Low Cracking Pressure**	This is the most compelling hypothesis. The fact that the **PRV is flowing at idle** (state s1) is highly anomalous and indicates it's opening at a much lower pressure than it should. A leaking PRV creates a permanent loss of pressure from the main system, perfectly explaining the major negative offset on **p_Combustion** (-11%). The control system's attempt to compensate for this leak by increasing pump output explains the positive offsets on **Flow_Pump** and **Flow_Bypass**.

2	**Combustion Chamber / Injector Nozzle(s)**	**Major Leak**	A major leak in the combustion chamber would explain the -11% pressure drop. However, it doesn't explain why the `Flow_PRV` was active at idle. A leaking PRV is a more direct cause that explains all symptoms, including the PRV flow itself.
3	**p_Combustion Sensor**	**Sensor Failure (Negative Offset)**	A failed sensor could explain the low `p_Combustion` reading, prompting the control system to increase pump and bypass flow. However, it doesn't explain the anomalous PRV flow at idle, which points to a real physical fault with the PRV.
4	**Main Fuel Pump**	**Over-performance**	This explains the high `Flow_Pump` and `Flow_Bypass` readings, but it cannot explain why the downstream combustion pressure is *low*. High flow should lead to high pressure.
5	**Bypass Valve**	**Incorrect Calibration**	The bypass valve is opening more to handle the excess flow from the compensating pump. This is a symptom, not the root cause.

3. Most Probable Hypothesis

The most probable root cause is that the **Pressure Relief Valve (PRV) is Stuck Partially Open or has a low cracking pressure.**

Reasoning: The "smoking gun" for this diagnosis is the `Flow_PRV` reading of +13.8% in state s1 (idle). The PRV should only ever activate during a severe over-pressure event, never at idle. This indicates the valve is faulty and is leaking fuel out of the main circuit from the very beginning of the run. This single, continuous leak provides a perfect explanation for all other symptoms: the system is constantly losing pressure (`p_Combustion` is low), and the control system is trying to compensate by working the pump harder (`Flow_Pump` and `Flow_Bypass` are high).